Chapter 10 | Computer Networks & Security

10.1 Data Communication Basics — Protocols, Modes and Signals

Protocol: a set of rules that governs how two computer systems interact — an **agreement on how communication components and DTEs are to communicate**, or in the third asked wording, **a set of formal operating rules, procedures or conventions governing a given process**. Its three elements are **syntax, semantics and timing**. A protocol is rules, never hardware — a CPU, tape or terminal is not a protocol — and **HTTP, DHCP and DNS are ALL used on the internet** (none is the odd one out).

Transmission mode	Direction	Asked examples
Simplex	One way only, always	**Keyboard → computer** (asked three times); computer → monitor
Half duplex	**Both ways, but only ONE way at a time**	Walkie-talkie; asked in both 'allows either direction but not simultaneously' wordings
Full duplex	**Both directions SIMULTANEOUSLY**	Telephone call; the mode simply called 'duplex' in two papers

Note: The fake options: **'half simplex' does not exist** (the valid pair in that list was simplex + duplex, I and III), and **multiplex is NOT a mode of data exchange** — multiplexing is the sharing of one medium by several signals, a different idea. **Parallel transmission = multiple pathways simultaneously** (a whole byte per clock; serial sends bits one by one). A **multiplexer combines many low-speed lines into one high-speed line**; the demultiplexer reverses it.

Bandwidth: the transmission capacity of a communication channel — how much data the link can carry per second, not how fast a signal travels.

Signals, Modulation and Encoding

A periodic analog signal (sine wave) is fully described by **peak amplitude, frequency and phase** — **wavelength is NOT one of the defining parameters** (it is derived). The wavelength formula:

$$\lambda = propagation\ speed \times period = propagation\ speed \div frequency$$

(1) Attenuation = the loss of energy as a signal propagates outward — asked again for fibre as **loss in signal power as light travels down the fiber**. Keep the trio apart: attenuation = loss of power, distortion = change of shape, noise = unwanted addition.

(2) Modulation figures: representing bits by changing the carrier's height = **amplitude modulation**; changing its frequency with constant amplitude = **frequency modulation** (the two figure-based 2021 questions).

(3) PCM (Pulse Code Modulation) converts an analog signal into a digital bit stream — sample, quantise, encode.

(4) Manchester encoding places a transition mid-bit, so pulse widths are **half** those of straight binary and the signalling rate doubles: a 10 Mbps token ring with differential Manchester runs at **20 Mbaud** (100 percent bandwidth overhead).

Shannon capacity bounds any channel. The asked working — 3 kHz bandwidth, signal-to-noise ratio 30 dB (a power ratio of 1000):

$$C = B\ log_2(1 + S/N) = 3000 \times log_2(1001) \approx 3000 \times 10 \approx 30\ kbps$$

Exam Points — 10.1

✦ Protocol = rules governing interaction (asked six ways); elements syntax + semantics + timing; never hardware; HTTP/DHCP/DNS all internet protocols.

✦ Simplex = keyboard→computer (×3); half duplex = one way at a time (×2); full duplex = both at once (×2); 'half simplex' fake; multiplex not a mode; parallel = multiple pathways; multiplexer merges low-speed lines.

✦ Bandwidth = channel capacity. Sine wave = amplitude/frequency/phase (wavelength NOT); λ = speed × period; attenuation = energy loss (×2); AM = height, FM = frequency; PCM analog→digital; Manchester half-width pulses, 10 Mbps → 20 Mbaud; Shannon worked ≈ 30 kbps.

10.2 Transmission Media

Class	Media	Anchor facts
Guided (wired)	Twisted pair, coaxial, **optical fibre**	Signal confined inside a physical conductor; fibre is guided (asked directly); **infrared is the non-guided intruder** in the guided list
Unguided (wireless)	Radio, microwave, infrared, light	**Transport electromagnetic waves without a physical conductor**, and ground propagation is a valid path (both statements true); **coaxial cable is the non-unguided intruder**; wireless transmission = radio + microwave + infrared ('more than one')

(1) **Twisted pair** carries data as **electric current**; the twist cancels interference; UTP terminates in an **RJ45** connector (8-pin Ethernet; RJ11 is the 4-pin telephone one).

(2) **Coaxial cable** is the guided medium **categorised by RG (Radio Government) ratings** — RG-59, RG-6 and so on.

(3) **Optical fibre**: the **core's refractive index is greater than the cladding's, so light is totally internally reflected into the core** — the same principle answered as 'works on total internal reflection'. Fibre offers the **highest transmission speed of any medium** (asked four times, wired and general) and, because it carries light not current, it is the right choice **where electrical interference is present**. Attenuation in fibre = the loss of light power along the run.

Exam Points — 10.2

✦ Guided = wired (fibre included); unguided = EM waves without conductor + ground propagation (both true); coaxial not unguided; infrared not guided; wireless = radio/microwave/infrared.

✦ Twisted pair = electric current + RJ45; coaxial = RG ratings; fibre = core index > cladding → total internal reflection (×2), highest speed (×4), immune to electrical interference.

10.3 Network Types, Architectures and Topologies

(1) **Connections**: a **point-to-point link is a DEDICATED link between two devices** — it is NOT time-shared (only statement I). A **multipoint connection** is shared by many stations — the topology requiring it is the **bus**.

Network	Reach	Asked facts
LAN	A floor, a building, one campus	'Floor of a building' and 'single building' wordings; the three-building CAT6 campus near Pune is **still a LAN**; LAN facts: **faster than a WAN + covers a limited area** (II and III — satellites belong to WANs)
MAN	A city	**Metropolitan Area Network** (full form asked); **a collection of LANs + carries data AND voice** (both statements true)
WAN	Country / world	**Covers a wide area + communicates via satellites and microwaves** (I and II; it is SLOWER than a LAN); two branches in **different countries = WAN**

(2) **Peer-to-peer**: every computer is equal — **the FALSE claim is that it has a dedicated server**; any peer can access any other (only statement I in the 2021 pair); good for small setups of about 10 machines. **Client/server**: the central controller is the **SERVER, all others are CLIENTS**. A **network operating system** is the OS that shares files across the network.

Topologies

Topology	Wiring	Asked facts
Bus	One shared **backbone cable**	'Single cable known as backbone' and 'single connection connects all devices' wordings; the KVS definition question ('shared transmission medium heard by all') = **Bus**; needs the **multipoint** connection; one break kills the network
Star	Each device → **central hub/controller**	'Requires a central controller'; 'each device has a DEDICATED POINT-TO-POINT link to the central controller'; statement check: central failure kills the network (true) but **adding computers does NOT increase communication time** (false)
Ring	Each device to two neighbours	Token passing; one of the standard five
Mesh	Every device to every other	**Highest reliability** (asked three times) and **highest number of ports + largest cabling** — n(n−1)/2 links

Note: **Fake topologies planted in the papers**: Car, Disk, Stack, Singleton — none exists (the real set: bus, star, ring, mesh, tree, hybrid). 'Which is a topology: star, bus, car?' → I and II only. If an option sounds like a data structure or a vehicle, it is not a topology.

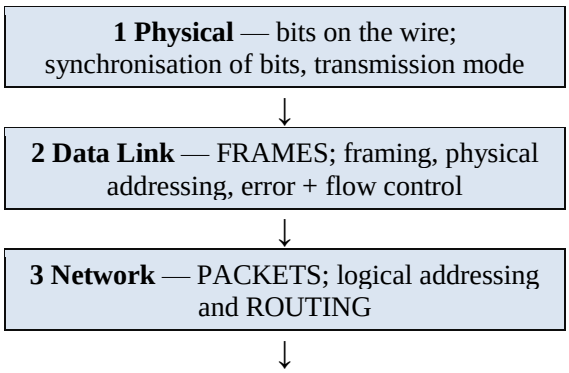
(3) **Switching**: **circuit switching requires a dedicated communication channel** — a **dedicated physical path through the intermediate nodes** held for the whole call (asked three times). **Packet switching** shares links — it is what **ARPANET used** and what **the internet works on**. Message switching stores and forwards whole messages.

Exam Points — 10.3

- ✦ Point-to-point = dedicated, not time-shared; multipoint → bus. LAN = building/floor/campus (faster + limited); MAN = city, collection of LANs, data + voice; WAN = wide + satellites, slower, different countries.
- ✦ P2P: no dedicated server (the false claim), equal peers; client/server = server; clients; network OS shares files.
- ✦ Bus = backbone/single cable/shared medium (×4); star = central controller + dedicated links to centre (adding nodes doesn't slow it); mesh = most reliable (×3) + most ports and cabling; fakes: car/disk/stack/singleton.
- ✦ Circuit switching = dedicated path (×3); ARPANET and the internet = packet switching.

10.4 The OSI and TCP/IP Reference Models

The seven OSI layers, bottom-up — with the data-unit name changing at each step:



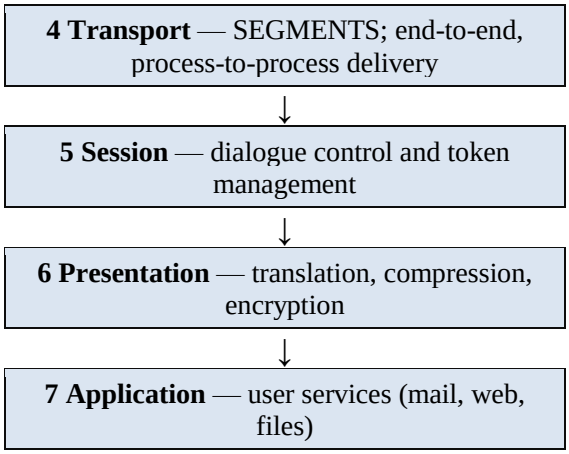


Fig. 10.1 — The OSI model. Layer five = Session (asked directly). Data units: frame → packet → segment going up.

(1) Layer responsibilities asked: session = **dialogue control + token management**; network = **routing** (asked five times across boards — 'determines the route', 'finding the best path'); presentation = translation and compression; physical = bit synchronisation. The RPSC matching: **A-2, B-4, C-1, D-3** (physical–bit sync, data link–error control and framing, session–dialog, presentation–translation). **NOT a network-layer function: error control and flow control** (data-link and transport jobs); the network layer does routing, intra-routing and congestion control.

(2) Chained vs end-to-end: layers **1–3 are chained (hop-by-hop, network support)**; layers **4–7 are end-to-end** — the 2021 statement pair reversed this, so BOTH claims were false. The **host-to-host (end-to-end) layers = Transport, Session, Presentation, Application**.

(3) Data link framing: the DLL **takes PACKETS from the NETWORK layer and encapsulates them into FRAMES**. In the data link layer, **error control is based on ARQ (retransmission)**; **flow control prevents data loss** at the transport layer by pacing the sender.

(4) Connection philosophy: in connectionless network service packets are **datagrams** travelling a **datagram network** (each routed independently); connection-oriented service = virtual circuit. The transport-layer statement triple — connectionless needs no setup, connection-oriented establishes first, **UDP is connectionless and unreliable** — all three TRUE.

TCP/IP layer	OSI equivalent	Asked facts
Application	Application + Presentation + Session	**RTP is an application-layer protocol** (with ICMP at the internet layer — the HTET pair); **RSVP is NOT an application-layer protocol** (it is signalling below)
Host-to-host	**Transport**	The layer corresponding to OSI transport (BPSC); holds TCP and UDP; **transport protocols = TCP, UDP and SCTP**; **end-to-end host-to-host connectivity = transport layer**
Internet	Network	Associated with **PACKETS**; ICMP lives here
Link	Data link + Physical	Frames and bits

(5) The TCP/IP protocol matching (2021): FTP–file transfer (iii), TFTP–trivial file transfer (iv), Telnet–remote login (i), UDP–send datagram (ii) → **a-iii, b-iv, c-i, d-ii**.

Devices by layer

Layer	Devices	Asked facts
Physical (1)	**Repeater**, hub, amplifier	Repeater regenerates the signal — **operates ONLY in the physical layer** (asked three times)

Data link (2)	**Bridge**, switch	**Bridge works at the data link layer** (asked three times — KVS twice, DSSSB once); **frames from one LAN to another travel via a BRIDGE**; a **switch connects devices within the same LAN** (forwards by MAC) and **prevents sniffing** on the segment
Network (3)	**Router**	Operates at the network layer (×2); **connects multiple networks together**; its **routing table tracks routes for forwarding data to destinations**
All layers	**Gateway**	**Connects systems using DIFFERENT protocols**; the device that **understands format and contents and TRANSLATES messages** (both KVS/HTET wordings)

Exam Points — 10.4

- ✦ Layer 5 = session (dialogue + tokens); network layer = routing (×5), NOT error/flow control; RPSC matching A-2,B-4,C-1,D-3; frame→packet→segment ladder; DLL frames the network layer's packets; DLL error control = ARQ.
- ✦ Layers 1–3 chained, 4–7 end-to-end (2021 pair: both false); host-to-host layers = T,S,P,A; datagrams → datagram network; UDP triple all true; internet layer = packets; host-to-host = OSI transport; TCP/UDP/SCTP; RTP app + ICMP internet; RSVP not app; FTP/TFTP/Telnet/UDP matching a-iii,b-iv,c-i,d-ii.
- ✦ Repeater L1 (×3); bridge L2 (×3) carries frames between LANs; switch = within a LAN, stops sniffers; router L3 joins networks, routing table = routes; gateway = different protocols + translates content (×2).

10.5 Data Link Layer — Framing, MAC, Error and Flow Control

(1) Framing marks: **bit stuffing distinguishes the beginning and ending FLAGS from the information** — an extra bit is stuffed whenever the data would imitate the flag (HDLC: a 0 after five 1s). The Ethernet **Start Frame Delimiter = 10101011** (after the 10101010 preamble). **Ethernet is the standard implemented by the NIC**; **10Base5 = Thick Ethernet** (500 m thick coax; 10Base2 = thin).

(2) Medium access: systems where users share a channel with possible conflicts = **contention** — **contention-based MAC = Ethernet's CSMA/CD**. On collision a station transmits the **jamming signal** so everyone backs off. **Token ring** is controlled access; its switches are housed in the **MAU (Multistation Access Unit)** — logically a ring, physically a star. **WDMA = Wavelength Division Multiple Access** (sharing a fibre by wavelength).

The CSMA/CD minimum-frame working (HTET): a frame must still be transmitting when the collision news returns, so transmission time ≥ 2 × propagation time:

$$Frame_{min} = 2 \times 25.6\,\mu s \times 10\,Mbps = 512\,bits = 64\,bytes$$

(3) Error control: **ARQ = the protocol in which the sender waits for a positive acknowledgement before advancing** to the next item. A **checksum detects errors or alterations in the transmitted message**; **CRC = Cyclic Redundancy Check**; and 'which is an error-detection code: Hamming, CRC, checksum?' → **more than one** (all three). The **residual error rate** measures lost or garbled messages as a fraction of those sent.

Hamming code

Parity bits sit at power-of-two positions; the code corrects ONE error and detects two. The parity-bit count must satisfy:

$$2^{r} \geq m + r + 1 \qquad (32\,data\,bits \Rightarrow r = 6, since\ 64 \geq 39)$$

(4) The Haryana statement triple: parity positions are powers of 2 (TRUE); Hamming corrects multiple errors (FALSE — single error only); data bits at even positions (FALSE — they occupy all non-power-of-two positions) → **only ONE statement correct**. And the capability question: **closer parity bits give LOWER error-correction capability** — correction grows with the code distance.

(5) Flow control = **the management of data flow between computers or between nodes** so a fast sender never swamps a slow receiver. **Sliding window beats stop-and-wait by allowing multiple packets before an acknowledgment** (higher throughput). **TCP handles flow control with the WINDOW SIZE field** (a zero window halts the sender). **X.25 specifies a DTE/DCE interface** for packet networks. **Sniffers are prevented by a SWITCHED network** — a switch forwards each frame only to its destination port.

Exam Points — 10.5

✦ Bit stuffing separates flags from data; SFD = 10101011; Ethernet = NIC standard; 10Base5 = thick Ethernet; contention → CSMA/CD → Ethernet (×2); jamming signal announces collisions; MAU houses token-ring switches; WDMA full form.

✦ CSMA/CD worked: 2 × 25.6 μs × 10 Mbps = 64 bytes. ARQ = positive-ack protocol; checksum detects alterations; CRC full form; Hamming/CRC/checksum all detection codes; residual error rate.

✦ Hamming: 2^r ≥ m+r+1 → 6 parity bits for 32 data; only 'powers of 2' statement true; closer parity bits = lower correction. Flow control paces sender; sliding window > stop-and-wait; TCP window size field; X.25 = DTE/DCE; switched network stops sniffers.

10.6 Network Layer — IP Addressing and Subnetting

(1) IPv4: a **32-bit** address of four octets, each 0–255 — so **153.254.287.83 and 256.10.12.5 are INVALID** (octets 287 and 256 overflow a byte; 255.255.255.10 and 240.x are fine). The 4-byte address divides into **network address + host address**. Class by first octet: **1–126 = A** (so **10.231.241.1 is Class A**), 128–191 = B, 192–223 = C, 224–239 = D (multicast).

(2) Subnetting: the **subnet mask identifies BOTH the network portion and the host portion** (the BPSC 'more than one' answer) — its 1s cover the network, its 0s the host. Subnetting allows **efficient use of IP addresses**. **CIDR = Classless Inter Domain Routing** (prefix of any length, written /n).

The asked subnet working — mask 255.255.240.0 leaves 12 host bits:

$$Hosts = 2^{12} - 2 = 4096 - 2 = 4094$$

(3) IPv4 header facts: **TTL (Time-to-Live) stops datagrams circulating forever** in routing loops (asked twice — decremented per hop, discarded at zero). The **fragmentation fields are Identification, Flags and Fragment Offset — TOS is NOT one**. Maximum datagram length = **65,535 bytes** (16-bit total-length field).

(4) IPv6: **128-bit** addresses (asked twice); fixed **40-byte** base header. In IPv4→IPv6 **header translation**, the mapped address takes the rightmost 32 bits and the priority is discarded, but **the flow label is NOT considered** — it has no IPv4 counterpart (the NOT-a-step answer).

(5) Address services: **DHCP allows BOTH manual and automatic IP assignment** — it is the **extension of the bootstrap protocol (BOOTP)**. **APIPA range = 169.254.0.1 to 169.254.255.254** (self-assigned when DHCP fails). **DNS converts a domain name into an IP address**; the **DNS client is the RESOLVER**.

Exam Points — 10.6

✦ IPv4 = 32 bits, network + host; octet > 255 invalid (287, 256); 10.x = Class A (ranges memorised); mask identifies both portions; subnetting = efficient addressing; CIDR full form; 255.255.240.0 → 4094 hosts.

✦ TTL kills looping packets (×2); fragmentation = ID/Flags/Offset (not TOS); max datagram 65535; IPv6 = 128 bits (×2), 40-byte header; translation drops the flow label; DHCP = BOOTP extension + manual and automatic; APIPA 169.254.0.1–169.254.255.254; DNS name→IP, client = resolver.

10.7 Routing, Switching and Congestion

Congestion: too many packets in (part of) the network causing delay and loss that degrade performance — the definition asked verbatim.

(1) Routing = determining the best path for data packets (asked four ways: 'primary function of routing', 'routing involves', 'what does the routing algorithm perform' → **decides the output line**, and 'routing table keeps track of' → **routes for forwarding data to destinations**). The **routing algorithm manages the routing tables and makes the routing decisions**.

(2) Adaptive vs non-adaptive: algorithms that do NOT base decisions on current traffic measurements = **non-adaptive** (static routing). **Flooding = sending each incoming packet on every outgoing line except the arrival line** — robust, wasteful, and a static algorithm.

(3) Distance vector vs link state: the Haryana triple — distance vector suffers **count-to-infinity** (TRUE); flooding is static (TRUE); link state is Bellman-Ford (FALSE — **link state = Dijkstra; distance vector = Bellman-Ford**) → **only two correct**. A distance-vector table entry holds the **preferred OUTGOING line + the estimated time** to the destination (RPSC key: output line, estimated time).

The hierarchical-routing working (Haryana): 630 routers, 3 levels, 7 clusters × 9 regions × 10 routers — each router lists its own region's routers, the other regions of its cluster, and the other clusters:

$$Entries = 10 + (9 - 1) + (7 - 1) = 10 + 8 + 6 = 24$$

(4) Virtual circuits: **circuit setup is required in a virtual-circuit network but not in a datagram network** — the direct 2023 statement. **DVMRP = Distance Vector Multicast Routing Protocol**. **GPRS offers end-to-end packet-switched transfer in BOTH point-to-point and point-to-multipoint** modes. The MANET attack that **injects false routing information to loop packets = the Byzantine attack**.

Exam Points — 10.7

✦ Congestion = too many packets → delay + loss. Routing = best path / output line / routing-table routes (×4); routing algorithm manages tables; non-adaptive ignores current traffic; flooding = every line but the arrival one.

✦ DV = Bellman-Ford + count-to-infinity; LS = Dijkstra (two of three statements correct); DV entry = output line + estimated time; hierarchical worked 10+8+6 = 24; VC needs setup, datagram doesn't; DVMRP full form; GPRS both modes; Byzantine attack loops packets.

10.8 Transport Layer — TCP, UDP, Ports and Connections

(1) Addressing ladder: the **transport-layer address is the PORT NUMBER** — it **specifies the communicating process on the two end systems**; transport delivers **process-to-process, between processes on different devices**, providing **end-to-end host-to-host connectivity**. A **socket = the endpoint of inter-process communication** — a **software interface for network communication**, concretely **IP address + port number** (asked three times). **Upward multiplexing = combining multiple data streams onto one connection**.

Property	TCP	UDP
Connection	**Connection-oriented** — the FALSE claim in the 2023 item was 'connection-less'; all of connection-oriented + 3-way handshake + single-stream input are TRUE (BPSC 'more than one')	**Connectionless, unreliable** — sends **encapsulated IP datagrams without establishing a connection**; supports **multiplexing via ports**
Setup	**Three-way handshake: SYN → SYN-ACK → ACK** (asked three times — the exchange that indicates successful establishment)	None — a **datagram network is a CONNECTIONLESS network** (×2)
Delivery	**In the order transmitted** — reliable ordered byte stream; **full-duplex**	Best effort
Header	**20 bytes without options**; window size field (flow control); sequence + acknowledgement numbers	**Four fields only: source port, destination port, length, checksum — NO sequence number** (8 bytes)

DSSSB – TGT COMPUTER SCIENCE (PRACTICE KING) BOOK

Call/Message: 7011121290 Available @ Amazon: https://amzn.in/d/07E04o9B

(2) **Acknowledgement number = the NEXT byte expected** (cumulative ACK, Haryana). **Handshaking** in general = the exchange of control signals where the receiver acknowledges receipt (KVS 2017).

(3) **Connection termination**: initiated by the **FIN flag**; a clean close **sends FIN and ACK sequentially in both directions**, so **both sender and receiver agree to close**. A received **RST packet terminates the connection IMMEDIATELY** (abrupt abort). The **retransmission timer** resends a segment whose acknowledgment never arrived.

(4) **Congestion control phases**: **slow start, congestion avoidance, fast retransmit** — **data fragmentation is NOT one**. In **slow start the congestion window approximately DOUBLES every round-trip time** (exponential; congestion avoidance then adds 1 MSS per RTT). Selective Repeat (a sliding-window ARQ) **buffers out-of-order frames until the missing ones arrive** — Go-Back-N would discard them.

(5) **Ports**: the 2023 matching — **7 Echo, 9 Discard, 11 SYSTAT, 13 Daytime** → a-ii, b-iii, c-iv, d-i. **IMAP = port 143** (the remote-mail daemon port asked in the OSI band). Memorise the ladder: 20/21 FTP, 25 SMTP, 53 DNS, 80 HTTP, 143 IMAP, 443 HTTPS.

(6) **SSL/TLS**: **TLS is the cryptographic protocol ensuring privacy in secure web communication** (HTTPS). SSL requires a server certificate, uses a dedicated socket and **operates ABOVE the transport layer — the FALSE characteristic is 'between the network and transport layer'**. What SSL does: private connection + encryption + safe transmission = **all of the above**.

Exam Points — 10.8

- ✦ Port = transport address = the process identifier; socket = IPC endpoint = IP + port (×3); process-to-process on different devices; end-to-end = transport; upward multiplexing merges streams.
- ✦ TCP: connection-oriented (false claim = connectionless), 3-way handshake SYN/SYN-ACK/ACK (×3), single input stream, ordered delivery, full-duplex, 20-byte header; ACK = next byte expected. UDP: connectionless + unreliable, datagrams without connection, multiplexing, header = ports + length + checksum (no sequence number).
- ✦ FIN starts termination; FIN + ACK both ways = agreed close; RST = immediate abort; retransmission timer resends. Congestion phases exclude fragmentation; slow start doubles per RTT; Selective Repeat buffers out-of-order frames.
- ✦ Ports 7/9/11/13 = Echo/Discard/SYSTAT/Daytime; IMAP 143. TLS = web privacy; SSL sits above transport (not between network and transport); SSL does all three listed jobs; datagram network = connectionless (×2).

10.9 Application Layer Protocols and Control Protocols

Protocol	Job — as asked	Notes
HTTP	**Web browsing** — Hypertext Transfer Protocol (full form asked twice)	**STATELESS** — keeps no memory between requests (cookies add state); runs on TCP port 80
FTP	**Uploading and downloading files** from remote computers (asked four times)	Control port 21, data port 20; **XMODEM** is the other file-transfer protocol asked (2021)
SMTP	**Sending e-mail** between machines (asked five times across boards)	A mail server **relaying mail onwards acts as an SMTP CLIENT** for that transfer; POP3/IMAP retrieve
DNS	Name → IP resolution	Client = **resolver**
Telnet	Remote login	From the TCP/IP matching
DHCP	**Manual + automatic IP assignment**; BOOTP extension	Also supplies the co-located care-of address in Mobile IP

(1) **Multimedia protocols**: **RTP carries real-time multimedia (audio and video)** — asked twice, once as 'basic transmission of multimedia content'. **RTSP is the remote control: it controls streaming servers,**

establishes/controls sessions AND controls playback ('more than one'); its **TEARDOWN state = the server breaks down the connection** and frees resources. **XMPP = presence detection, contact management and real-time messaging** in IM.

(2) **Web and mail extras**: **AJAX reduces bandwidth by sending and retrieving only small amounts of data** rather than reloading pages. **MIME = Multipurpose Internet Mail Extension** (non-ASCII mail content). **X.12 is the EDI protocol** — electronic data interchange between businesses. **E-mail is the application NOT needing high-speed equipment** (small and delay-tolerant). An **application gateway filters application-specific commands such as http:get and http:post** — the firewall type that reads the application protocol.

Exam Points — 10.9

✦ HTTP = web, full form (×2), stateless; FTP = files (×4), 21/20, XMODEM too; SMTP = mail (×5), relaying server = SMTP client; DNS resolver; DHCP manual + automatic.

✦ RTP = real-time A/V (×2); RTSP = all three control roles, TEARDOWN breaks the connection; XMPP = presence + IM; AJAX = small data transfers; MIME full form; X.12 = EDI; e-mail needs no high-speed link; application gateway filters http:get/post.

10.10 Wireless, Mobile and Cellular Networks

(1) **Wi-Fi standards**: **IEEE 802.11 = wireless LAN** (802.3 is wired Ethernet); **802.11i = the WLAN SECURITY standard** (source of WPA2); **WPA2 secures Wi-Fi**; **802.1x serves BOTH authentication AND master-session-key generation**. In an infrastructure 802.11 network, the **BSA (Basic Service Area) is simply a CELL**. **MIMO (Multiple Input Multiple Output) enhances Wi-Fi range and reliability** with multiple antennas.

(2) **Cellular access and generations**: **TDMA shares one frequency channel by TIME SLOTS**; **CDMA is 2G**; speed order **GSM < GPRS < EDGE < LTE** (ascending, asked directly). **GSM = Global System for Mobile communication**; its architecture = **Radio Subsystem + Networking and Switching Subsystem + Operation Support Subsystem (all three)**. **GPRS = General Packet Radio Service** — packet data on GSM. The **two basic cellular systems = circuit-switched and packet-switched**.

(3) **Mobility machinery**: **handoff** = the base station transfers a moving call to the cell with the strongest signal; **inter-system handover** = satellite → terrestrial transfer. **BTS = Base Transceiver Station**. **The VLR is responsible for informing the HLR of a mobile station's location changes**. Location-update schemes: periodic, movement-based, distance-based — **'cost-based' does NOT exist**.

(4) **Mobile IP**: the mobile node keeps its home address; the **care-of address (COA) solves the mobile-IP problem** — it is **the address of the current tunnel endpoint, obtainable via DHCP** (co-located COA); the **foreign agent is the router in the foreign network the mobile node currently attaches to**. **IPSec tunnel mode protects the ENTIRE packet** (transport mode: payload only).

(5) **Mobile computing corner**: **mobile agents = TRANSPORTABLE agents** (pre-defined or undefined path); a **Wi-Fi router is the fixed-and-wireless device** (desktop = fixed wired; laptop/dongle = portable); mobile OS list — Windows Mobile, Palm OS, Symbian — **'Manet OS' is fake** (a MANET is an ad-hoc network). **Limitations of mobile computing = power consumption, human interface, range/bandwidth, interference (ii, iii, iv, v)** — mobility and wireless connectivity are benefits. The mobility matching: **macro = global network, micro = single administrative domain, ad hoc = MANET (a-ii, b-iii, c-i)**. **Bluetooth 5.0 = 2 Mbps maximum** (double 4.x, quadruple range); **Bluetooth is the usual short-range IoT link**; **MMID identifies mobile money**.

Exam Points — 10.10

✦ 802.11 = WLAN; 802.11i = WLAN security; WPA2 = Wi-Fi; 802.1x = authentication + MSK (both); BSA = a cell; MIMO boosts range; TDMA = time slots; CDMA = 2G; GSM<GPRS<EDGE<LTE.

✦ GSM full form + three subsystems (all); GPRS full form; cellular = circuit + packet switched; handoff = strongest-signal transfer; inter-system = satellite→terrestrial; BTS full form; VLR informs HLR; no cost-based update scheme.

✦ Mobile IP: COA = solution + tunnel endpoint via DHCP; foreign agent = router in visited network; IPSec tunnel mode = whole packet. Agents = transportable; Wi-Fi router = fixed + wireless; Manet OS fake; limitations ii,iii,iv,v; mobility matching a-ii,b-iii,c-i; Bluetooth 5.0 = 2 Mbps + IoT; MMID = mobile money.

10.11 Cryptography, Keys and Digital Signatures

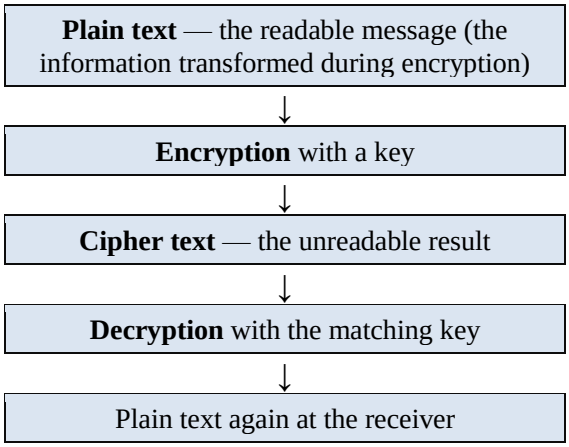


Fig. 10.2 — The encryption pipeline. Cryptanalysis = the process of breaking cipher text to obtain the plain text without the key.

Scheme	Keys	Asked facts
Symmetric	**One shared key encrypts AND decrypts**	AES, RC4, DES, IDEA, 3DES; a temporary per-conversation key = the **SESSION key**
Asymmetric (public key)	**Different keys: public + private pair**	**Public-key cryptography IS asymmetric cryptography**; **RSA is the non-symmetric one** in the AES/RC4/RSA/DES list; **the RECEIVER keeps the private key**; a message encrypted with the recipient's public key is decrypted with **the recipient's PRIVATE key**

(1) Key sizes (all asked): DES **56 bits**; **Triple DES 168 bits**; **IDEA 128 bits** (64-bit blocks). A **transposition cipher reorders the letters but does not disguise them** (substitution replaces them). The **man-in-the-middle attack on Diffie-Hellman = the BUCKET BRIGADE attack**.

(2) Digital signatures: sign with the **sender's private key**, verify with the **sender's PUBLIC key**; integrity is guaranteed by the **HASH VALUE of the original message** — if the recomputed digest matches, nothing changed.

(3) Digital certificates: they **verify identity using PKI**; contents = **public key + owner information + validity period** (never the private key). The 2021 statement triple: the certificate DOES name its CA (I false), **certificates are based on public-key cryptography (II TRUE — the only correct one)**, and they come in several strengths, not only 128-bit (III false).

(4) IPSec operates in TRANSPORT and TUNNEL modes (asked directly; tunnel = whole packet, transport = payload). And the hash-function stray: methods of OBTAINING a hash = folding, truncation, modular arithmetic — **random probing is collision resolution, not a hash method**.

Exam Points — 10.11

✦ Plain → cipher → plain; encryption input = plain text; cryptanalysis breaks ciphers. Symmetric = one key (AES/RC4/DES); asymmetric = public + private = public-key crypto; RSA asymmetric; receiver holds the private key; decrypt with recipient's private key; session key per conversation.

✦ DES 56, 3DES 168, IDEA 128; transposition reorders without disguising; Diffie-Hellman MITM = bucket brigade. Signatures: sign private, verify with sender's public; integrity = message hash. Certificates: PKI identity, public key + owner + validity, only 'based on public-key crypto' true. IPSec = transport + tunnel; random probing not a hash method.

10.12 Network Security — Goals, Firewalls, Attacks and Malware

(1) **The three security goals (CIA triad) = Confidentiality, Integrity and Availability**. **Nonrepudiation is the service that protects against repudiation** — a party cannot later deny sending a message. **Authentication = the process of verifying a user's identity** (asked twice; authorization then decides what they may do).

(2) **Firewalls**: a firewall is **a system designed to prevent unauthorized access** between networks. The **PROXY firewall acts as an intermediary for requests between a client and a server**; the **application gateway** filters application commands (http:get/post, 10.9). The four types to memorise: packet filter (headers), stateful (connections), circuit-level gateway (sessions), proxy/application gateway (content).

(3) **Intrusion detection and decoys**: **signature-based detection is the MOST COMMON IDS method** (matches known attack patterns; anomaly-based flags deviations). A **stack-based IDS is integrated closely with the TCP/IP stack and watches packets** as they climb the layers. A **honeypot is a decoy system that attracts attackers so their tactics can be learned**.

(4) **Attack vocabulary**: **spoofing = disguising a communication from an unknown source as being from a trusted source**; **masquerade = one entity pretending to be another**; DoS attacks include **Teardrop, Smurf and SYN flood — Code Red is NOT a DoS attack, it is a worm**. **Hacking is NOT a physical threat** (storm, fire and flood are); external threats include front-door, back-door and DoS — **'underground threats' do not exist**. A **keylogger** captures typed passwords — defeated by the on-screen **virtual keyboard** in net banking.

(5) **Viruses**: a virus is code that **attaches itself to a legitimate program and runs when that program runs**; its essential abilities — **clone itself AND damage data or programs ('more than one')**. A virus corrupting the **partition table makes the hard disk totally inaccessible** (corrupting the FAT only loses files). Odd-one-out: **Code Red, Melissa and I Love You are malware; COMODO is an antivirus** — the odd one.

Exam Points — 10.12

✦ CIA = confidentiality/integrity/availability; nonrepudiation blocks denial; authentication = identity check (×2). Firewall = prevents unauthorized access; proxy firewall = client-server intermediary; signature IDS most common; stack IDS inside TCP/IP; honeypot = decoy.

✦ Spoofing = fake trusted source; masquerade = pretend identity; Code Red = worm, not DoS (Teardrop/Smurf/SYN flood are); hacking not physical; no 'underground' threat; keylogger vs virtual keyboard; virus = clones + damages ('more than one'), attaches to programs; partition-table corruption = disk gone; Comodo the antivirus odd-one-out.

10.13 Internet, WWW and E-commerce

(1) **The internet is a VAST COLLECTION OF DIFFERENT NETWORKS** — not a single network. Its origin: **ARPANET, the first network that initiated the internet** (1969, and the first to use packet switching — both facts asked). To join, a computer **must connect to an Internet Service Provider (ISP)**; ISPs **exchange traffic between their networks at Internet Exchange Points**. **DSL (Digital Subscriber Line) provides internet access over the wires of the local telephone network**. The essential components for a computer to communicate over the internet: **NIC and TCP/IP**; and world-wide networks are interconnected chiefly **for messages for communication**.

(2) **VPN = a private network that uses a public network to connect remote sites or users** — asked twice (also as 'extends a private network across public networks'); the traffic is tunnelled and encrypted.

(3) **Web oddments**: a **CAPTCHA is the picture-format text you retype to prove you are human**. **Elastic traffic adjusts over wide ranges of delay and throughput and still meets the application's needs** (file transfer, mail;

real-time audio/video is inelastic). **E-commerce = buying and selling goods over the Internet**; a **REVERSE auction** is where the buyer posts a need and **sellers bid successively LOWER until one remains**.

(4) **The two HTET matchings**: Bing = search engine, Flipkart = online shopping, Facebook = social networking, Chrome = web browser, gmail = e-mail → **I-C, II-D, III-A, IV-E, V-B**. Likewise 3D Studio = animation, Ubuntu = OS, DuckDuckGo = search engine, Maxthon = web browser, Vipre Rescue = antivirus → **I-C, II-D, III-A, IV-E, V-B** (same letter pattern). Method: fix the two pairs you are surest of and eliminate.

Stray (Chapter 12 topics): Two software-engineering/MIS items appear in this chapter's papers and are covered in Chapter 12: a **PERT chart is a network of boxes and arrows where boxes = activities and arrows = task dependencies (BOTH statements true)**; and **IRCTC reservation, electronic banking and logistics management are ALL correct examples of Management Information Systems (I, II and III)**.

Exam Points — 10.13

✦ Internet = vast collection of different networks; born from ARPANET (first network, packet switched); join via ISP; ISPs peer at Internet Exchange Points; DSL = digital data over telephone wires; essentials = NIC + TCP/IP; interconnection purpose = messages.

✦ VPN = private network over a public one (×2); CAPTCHA = picture text proving human; elastic traffic adjusts to delay/throughput; e-commerce = buying/selling online; reverse auction = sellers bid down; both HTET matchings = I-C, II-D, III-A, IV-E, V-B.

Quick Revision — Chapter 10 (35 Points)

1. Protocol = rules governing how systems interact (syntax + semantics + timing); never hardware; HTTP/DHCP/DNS all internet protocols.
2. Modes: simplex (keyboard→computer), half duplex (one way at a time), full duplex (both at once); 'half simplex' and 'multiplex' are not modes; parallel = multiple pathways simultaneously.
3. Sine wave = amplitude + frequency + phase (NOT wavelength); λ = speed × period; attenuation = loss of signal energy/power.
4. AM changes carrier height, FM changes frequency; PCM digitises analog; Manchester = half-width pulses, 10 Mbps → 20 Mbaud; Shannon: 3 kHz at 30 dB ≈ 30 kbps.
5. Guided = twisted pair + coaxial + fibre; unguided = radio + microwave + infrared (EM waves, no conductor); multiplexer merges low-speed lines into one.
6. Twisted pair = electric current, RJ45; coaxial = RG ratings; fibre = total internal reflection (core index > cladding), fastest medium, immune to electrical interference.
7. Point-to-point = dedicated link (not time-shared); multipoint → bus topology.
8. LAN = building/floor/campus, faster + limited; MAN = city, collection of LANs, data + voice; WAN = wide area, satellites/microwaves, slower, different countries.
9. P2P = no dedicated server, equal peers; client/server = server (controller) + clients; NOS shares files across a network.
10. Bus = one backbone cable, shared by all; star = dedicated links to a central controller (adding nodes doesn't slow it); mesh = most reliable, most ports/cabling; fakes: car, disk, stack, singleton.
11. Circuit switching = dedicated physical path (phone); packet switching = ARPANET + internet.
12. OSI: Physical, Data link, Network, Transport, Session (layer 5, dialogue + tokens), Presentation (translation), Application; frame → packet → segment.
13. Network layer = routing (NOT error/flow control); layers 1–3 chained, 4–7 end-to-end; host-to-host layers = T, S, P, A.
14. TCP/IP: internet layer carries packets; host-to-host layer = OSI transport; transport protocols = TCP + UDP + SCTP; RTP app layer, ICMP internet layer; RSVP not application.

15. Devices: repeater L1, bridge + switch L2 (bridge joins LANs, switch stops sniffers), router L3 (joins networks), gateway translates between different protocols.
16. Bit stuffing separates flags from data; SFD = 10101011; 10Base5 = thick Ethernet; MAU houses token-ring switching.
17. CSMA/CD = Ethernet contention; jamming signal announces a collision; minimum frame = 2 × propagation × bandwidth = 64 bytes at 10 Mbps/25.6 μs.
18. ARQ = wait for positive ack, retransmit; checksum/CRC/Hamming all detect errors; CRC = Cyclic Redundancy Check.
19. Hamming: $2^r \geq m + r + 1$ (32 data bits → 6 parity); parity at powers of 2; corrects ONE error; closer parity bits = lower correction capability.
20. Flow control paces the sender (TCP: window size field); sliding window beats stop-and-wait; X.25 = DTE/DCE interface; residual error rate = lost/garbled fraction.
21. IPv4 = 32 bits, network + host, octets 0–255 (287/256 invalid); class A = 1–126 (10.x.x.x); subnet mask marks both portions; 255.255.240.0 → 4094 hosts; CIDR = classless.
22. TTL prevents endless loops; fragmentation fields = ID + Flags + Offset (not TOS); max datagram 65,535 B; IPv6 = 128 bits, 40-byte header, flow label dropped in translation.
23. DHCP = BOOTP extension, manual + automatic assignment; APIPA = 169.254.0.1–169.254.255.254; DNS = name→IP, client = resolver.
24. Congestion = too many packets → delay + loss; routing = choosing the best path/output line; non-adaptive ignores current traffic; flooding = every line except arrival.
25. Distance vector = Bellman-Ford (count-to-infinity; table: output line + estimated time); link state = Dijkstra; hierarchical 630-router sum = 10 + 8 + 6 = 24 entries.
26. Port = transport address (identifies the process); socket = IP + port, the IPC endpoint; upward multiplexing = many streams on one connection.
27. TCP: connection-oriented, 3-way handshake (SYN/SYN-ACK/ACK), ordered full-duplex byte stream, 20-byte header, ACK = next byte expected; UDP: connectionless, 8-byte header, no sequence number.
28. FIN opens a mutually-agreed close (FIN + ACK both ways); RST kills the connection at once; retransmission timer resends; slow start doubles the window per RTT; Selective Repeat buffers out-of-order frames.
29. Ports: 7 Echo, 9 Discard, 11 SYSTAT, 13 Daytime; 21 FTP, 25 SMTP, 53 DNS, 80 HTTP, 143 IMAP; SSL/TLS sits above transport and secures HTTPS.
30. HTTP = stateless web protocol; FTP = files; SMTP = mail (sender acts as SMTP client); RTP = real-time A/V; RTSP controls streaming (TEARDOWN ends it); MIME extends mail; X.12 = EDI; AJAX sends only small data.
31. 802.11 = WLAN, 802.11i = its security (WPA2), 802.1x = authentication + keys; BSA = a cell; GSM < GPRS < EDGE < LTE; Bluetooth 5.0 = 2 Mbps, the IoT short-range link.
32. Mobile IP: care-of address = tunnel endpoint (via DHCP), foreign agent = router in the visited network; VLR informs HLR; handoff follows the strongest signal.
33. Crypto: symmetric = one key (AES/RC4/DES/3DES-168/IDEA-128); asymmetric = RSA, receiver keeps the private key; sign with private, verify with public; hash guarantees integrity; certificates = PKI (public key + owner + validity); Diffie-Hellman MITM = bucket brigade; IPSec = transport + tunnel modes.
34. Security: CIA triad; authentication verifies identity; firewall blocks unauthorized access (proxy = intermediary); signature IDS most common; honeypot = decoy; spoofing/masquerade fake identity; Code Red = worm not DoS; virus clones itself + damages, partition-table hit = disk inaccessible.
35. Internet = collection of networks from ARPANET, joined via ISPs peering at IXPs; DSL over phone wires; VPN = private over public; CAPTCHA proves humanity; elastic traffic adapts; reverse auction = sellers bid down.

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